Meysam Varasteh

Ph.D. Student in Computer Science | Recommender Systems & Interpretable Machine Learning

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Boulder, Colorado, USA

Research Interests

Recommender systems; interpretable and controllable machine learning; mechanistic interpretability, personalization and ranking

Education

University of Colorado Boulder **Aug. 2023 – Expected May 2028**

Ph.D. in Computer Science, Boulder, Colorado

- Advisor: Prof. Robin Burke.
- Thesis: Explainable AI in multi-stakeholder recommender systems

University of Tehran **2018 – 2021**
M.Sc. in Electrical and Computer Engineering, Tehran, Iran

K.N.Toosi University **2014 – 2017**
M.Sc. in Physics, Tehran, Iran

Petroleum University of Technology **2010 – 2014**
B.Sc. Electrical and Computer Engineering, Iran

Research Experience

Graduate Research Assistant **2023 – Present**
University of Colorado Boulder

Research Assistant **2021-2022**
Polytechnic University of Bari

Graduate Research Assistant **2018-2021**
University of Tehran

Graduate Research Assistant **2015-2018**
K.N.Toosi University

Publications and Manuscripts

Manuscript in preparation

1. **Meysam Varasteh** et al. "From Tokens to Concepts: A Post-hoc Concept Bottleneck for Interpretable and Steerable Recommendation." Manuscript in preparation.

Peer-reviewed papers

2. **Meysam Varasteh**, Robin Burke. "Why Didn't More People See It? Recommendation Transparency for Providers." *IntRS Workshop @ ACM RecSys 2026.* [paper]
3. **Meysam Varasteh**, Veronika Bogina, Noam Koenigstein, Robin Burke. "Why This, Not That? Mining User Profiles for Pair-wise Counterfactuals." *IntRS Workshop @ ACM RecSys 2026.* [paper]
4. **Meysam Varasteh**, Elizabeth McKinnie, Amanda Aird, Daniel Acuña, Robin Burke. "Comparative Explanations for Recommendation: Research Directions." *11th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS @ RecSys 2024)*, 2024, pp. 1–14. [paper]
5. **Meysam Varasteh**, Mehdi Soleiman Nejad, Hadi Moradi, Mohammad Amin Sadeghi, Ahmad Kalhor. "An Improved Hybrid Recommender System: Integrating Document Context-Based and Behavior-Based Methods." *31st International Conference on Electrical Engineering (ICEE)*, 2023. [IEEE]

Preprints and research manuscripts

6. **Meysam Varasteh**, Elizabeth McKinnie, Robin Burke. "Counterfactual Profile Perturbations for Recommendation Explanations: A Reproducibility Study" 2026
7. Mehdi Soleiman Nejad, **Meysam Varasteh**, Hadi Moradi, Mohammad Amin Sadeghi. "Designing a Sequential Recommendation System for Heterogeneous Interactions Using Transformers." *arXiv preprint arXiv:2205.00265*, 2022. [arXiv]
8. Yashar Deldjoo, **Meysam Varasteh**, Yara M. Bahram, Marco Antonio Insabato, Tommaso Di Noia. "Fairness Auditing Through the Lens of Counterfactual Explanations." Research manuscript / project on counterfactual fairness auditing.

Teaching Experience

Teaching Assistant – CSCI 5434: Probability for Computer Science **Spring 2024 - 2026**
University of Colorado Boulder

Teaching Assistant – CSCI 2400: Computer Systems **Fall 2023-2026**
University of Colorado Boulder

Teaching Assistant – Neural Networks and Deep Learning **Fall 2018-2020**
University of Tehran

Teaching Assistant – Statistical Inference **Spring 2019-2020**
University of Tehran

Teaching Assistant – Machine Learning **Spring 2018-2021**
University of Tehran

Mentorship and Service

Mentor for Undergraduate Mentees – CSCI 4830 **2026**
University of Colorado Boulder

Student Volunteer **2024**
ACM SIGIR

Student Volunteer **2023**
ACM RecSys

Data Mining Workshop Organizer **2022**
University of Tehran

Honors and Awards

2026 **Outstanding Teaching Assistant Award**, Department of Computer Science, University of Colorado Boulder.
2026 **WSDM NSF Conference Travel Grant**, National Science Foundation.
2025 **Departmental Summer Research Fellowship**, Department of Computer Science, University of Colorado Boulder.
2024 **SIGIR Conference Student Travel Grant**, ACM SIGIR.
2023 **Dean's and Departmental Excellence Fellowship**, University of Colorado Boulder.
2023 **Early Career Professional Development Fellowship**, University of Colorado Boulder.

Industry Experience

Data Scientist **Aug. 2018 – Feb. 2020**
Maktabkhooneh – Online Learning Platform

- Designed a sequential recommender system to personalize educational content recommendations from learners' historical interactions.

Technical Skills

Programming	Python, C++, JavaScript
ML/Data	PyTorch, TensorFlow, Hugging Face, scikit-learn, CUDA
Tools	Git, Docker, Podman, Linux, LaTeX
Research	RecBole, experimental evaluation, reproducible ML pipelines